1.6 $\lambda((a,b))=b-a$

proof: $\because \lambda((a,b-\varepsilon])\le\lambda((a,b))\le\lambda((a,b]),\ \forall\varepsilon\in(0,b-a)$

$\therefore b-a-\varepsilon\le\lambda((a,b))\le b-a$

$\therefore \lambda((a,b))=b-a$

0.1 $\alpha\sim y$, $x,y\in R$, if $y-x\in Q \Rightarrow \sim$ is equivalence relation.

0.2 $\Lambda=R/_\sim \Rightarrow \Lambda$ not countable

proof: $\because$ if Λ countable, $\because [x]$ countable $\therefore R$ countable

0.3 $\Omega:=\{$choose just one point $x\in\alpha, x\in(0,1)\mid \forall\alpha\in\Lambda\}$

($\because [x]\cap(0,1)\neq\emptyset, \forall x$)

0.3.3 $\Omega\subseteq(0,1)$

0.3.4 if $x,y\in\Omega, x\neq y \Rightarrow x\not\sim y$

0.4 $\Omega+p=\Omega+q$, or $(\Omega+p)\cap(\Omega+q)=\emptyset, \forall p,q\in Q$.

proof: If $(\Omega+p)\cap(\Omega+q)\neq\emptyset$

let $x\in(\Omega+p)\cap(\Omega+q)$ $\therefore x=\alpha+p, \alpha\in\Omega$,
$=\beta+q, \beta\in\Omega$.

$\therefore \alpha-\beta=q-p\in\Omega$, $\therefore \alpha\in[\beta]$, $\therefore \alpha=\beta$ (by 0.3.4)

$\therefore p=q$ $\therefore \Omega+p=\Omega+q$ ∎

λ#

1.

Goal: Try to define λ, $\Rightarrow \lambda\#$ $\because$ We remove $P(R)$

A2 (1) $\lambda((a,b])=b-a$,

(0) try to define λ, $\lambda: P(R)\to R^+\cup\{+\infty\}$,

here $P(R)$: all subset of R, $R^+:=\{x\mid x\ge 0\}$

λ extends the idea of length of interval

Properties λ should have:

(2) $\forall A\subseteq R, \forall x\in R, \lambda(A+x)=\lambda(A)$, here $A+x:=\{x+y\mid y\in A\}$

(3) $A=\bigcup_{j\ge1}A_j$, $A_j\cap A_k=\emptyset, \forall j\neq k \Rightarrow \lambda(A)=\sum_{j\ge1}\lambda(A_j)$

1.4 $\lambda(\emptyset)=0$

proof: $\because \lambda((-1,1])=\lambda((-1,1])+\lambda(\emptyset)+\lambda(\emptyset)+\cdots$

$\therefore \lambda(\emptyset)=0$

1.5 $E\subseteq F \Rightarrow \lambda(E)\le\lambda(F)$

proof: $\because F=E\cup(F\setminus E)$

$\therefore \lambda(F)=\lambda(E\cup(F\setminus E))=\lambda(E)+\lambda(F\setminus E)+\lambda(\emptyset)+\lambda(\emptyset)+\cdots$
$=\lambda(E)+\lambda(F\setminus E)$

$\therefore \lambda(E)\le\lambda(F)$

$\therefore \lambda(\Omega)=0$ ($\because$ if $\lambda(\Omega)>0, \Rightarrow \sum_{\substack{q\in Q\\ -1<q<1}} \lambda(\Omega)=+\infty$)

2.4 $\lambda\left(\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)\right)=0$

proof: $\because \lambda(\Omega+q)=\lambda(\Omega)=0$

$\therefore \lambda(\Sigma(\Omega+q))=\Sigma\lambda(q+\Omega)=0$

2.5 $\lambda \not\equiv$

proof: $\because 1=\lambda((0,1)) \leq \lambda\left(\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)\right)=0$ (by 0.9) (by 2.4)

$\therefore \lambda \not\equiv$.

0.5 $p\neq q,\ p,q\in Q \Rightarrow (\Omega+p)\cap(\Omega+q)=\phi$ (by 0.4)

0.6 $\Omega+q\subseteq(-1,2),\ q\in Q,\ q\in(-1,1)$

($\because \Omega\subseteq(0,1)$)

0.7.2 $\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)$ disjoint (by 0.5)

0.7.3 $\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)\subseteq(-1,2)$ (from 0.6)

0.8 $x\in(0,1) \Rightarrow \exists\, q\in Q,\ q\in(-1,1),\ x\in\Omega+q$

proof: $\because \exists\, \alpha\in[x]\cap\Omega,\ \alpha\in(0,1)$ 且 $\because x-\alpha=q\in Q$

$\therefore q=x-\alpha\in(-1,1)\ ;\ x=\alpha+q\in\Omega+q$

0.9 $(0,1)\subseteq\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)$ (by 0.8)

2. proof of 1.

2.2 $\lambda\left(\sum_{\substack{q\in Q\\ -1<q<1}}(\Omega+q)\right)\leq\lambda((-1,2))=3$ (from 1.5, 1.6, 0.7.3)

2.3 $\lambda(\Omega)=0$

proof: $\because \sum_{\substack{q\in Q\\ -1<q<1}}\lambda(q+\Omega)=\sum_{\substack{q\in Q\\ -1<q<1}}\lambda(\Omega)\leq 3$ (by 0.7.2, 1.2, 2.2)